

Chia-Yuan Chiang

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EDUCATION

National Central University

B.S. in Electrical Engineering

Taoyuan, Taiwan

Sep. 2024 – Sep. 2027 (Expected)

Ming Chi University of Technology

B.S. in Electrical Engineering (Transferred)

New Taipei, Taiwan

Sep. 2023 – Jun. 2024

Nangang Senior High School

Electronics Technology

Taipei, Taiwan

Sep. 2020 – Jun. 2023

EXPERIENCE

Undergraduate Researcher

May 2025 – Present

Prof. Shu-Yen Lin's Lab, National Central University

Taoyuan, Taiwan

- Led an early-warning model for ICU Ventilator-Associated Pneumonia (VAP); designed a stay-level 5-fold cross-validation scheme to eliminate patient-level data leakage present in the public benchmark.
- Applied Integrated Gradients feature attribution to reduce the model to 4 non-invasive features; the resulting LSTM reached 0.98–0.99 AUROC at a 72-hour prediction horizon.
- Manuscript submitted to IEEE GCCE 2026.

Engineering Advisor & Systems Integrator

2023 – Present

FRC Team 7645

Taipei, Taiwan

- Own robot system architecture: multi-sensor fusion, PID closed-loop motor control tuning, and software/hardware interface definitions.
- Introduced LLM-assisted tooling that cut iterative firmware debugging time by roughly one week; trained junior members to debug from root-cause analysis instead of trial-and-error.
- Rebuilt the team website, reducing repetitive PR/outreach inquiries.

Competitor

Jun. 2020 – Sep. 2023

Nangang Senior High School – NK-MTC 7645

Taipei, Taiwan

- Generalist across LabVIEW/Arduino firmware, FRC electrical wiring, AutoCAD modeling, and 3D-printed mechanical parts; competed in 3 FRC regional events and represented the team at the National Skills Competition.

PROJECTS

PCB Defect Detection | *Azure Custom Vision, Python*

2025

- Trained an object-detection model to localize PCB defects; used 600px defect-centered tiling to raise defect bounding-box area share from ~2% to ~12% of each tile.
- Validated on held-out board types (06/07) unseen during training; reached 100% precision, 69.7% recall, 81.9% mAP at 0.5 threshold.

Job Radar | *Python, Playwright, GitHub Actions, Cloudflare Pages*

2026 – Present

- Built a personal job-listing aggregator scraping 104, Cake, and LinkedIn (~1,300 listings/day) via a daily GitHub Actions cron; reverse-engineered 104's internal SPA API after the legacy API was deprecated.
- Designed a config-driven scoring engine (role fit, location commute, experience threshold) and a static-site dashboard build pipeline with no backend or database.

AWS x BitoPro Hackathon – AI Fraud Detection

2026

- Built an account risk-scoring dashboard (AUC 83.2%) with 5-fold CV and decision-threshold tuning (optimal 0.24) over 12,753 user accounts.

TECHNICAL SKILLS

Languages & Firmware: C/C++ (STM32/Cortex-M), Python, QMK/ChibiOS HAL, LabVIEW, Arduino/ESP32, UART/SPI/I2C

Machine Learning: PyTorch, TensorFlow/Keras, LSTM/Time-Series, Feature Engineering, Signal Processing, SHAP/XAI, TinyML

EDA / Hardware: Altium Designer, KiCAD, OrCAD, EasyEDA Pro

Manufacturing: CNC Milling, Mastercam, 3D Printing, Laser Cutting, Welding, AutoCAD, Fusion 360, Autodesk Inventor

AWARDS & CERTIFICATIONS

2022 FRC Taiwan Hon Hai Regional – Finalist Alliance & Innovation in Control Award · Siemens SMS-CP Level 1 Mechatronics Certification (2022) · 52nd National Skills Competition – Team Project Category Representative (2022) · 2020 FRC CIT 5G Regional – Excellence in Engineering (Industrial Design) Award